

Adaptive Evidence Acquisition for Grounded Video Question Answering

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Final Presentation

Core issue

VideoQA systems often answer from a fixed amount of visual context, even when the answer depends on a short event.

Research question

How much visual evidence is enough to answer a video question while staying temporally grounded?

This creates two problems:

- computation is spent on irrelevant frames;
- a correct answer may not be grounded in the supporting moment.

What prior work gives us

- VideoQA benchmarks: MovieQA, TGIF-QA, TVQA, NExT-QA.
- Vision-language pretraining: CLIP, VideoCLIP, CLIP4Clip.
- Grounded VideoQA: NExT-GQA asks whether answers are supported by visual evidence.

Gap we study

- Most systems use fixed context or fixed top- K retrieval.
- Retrieval quality, answer accuracy, and grounding are often evaluated separately.
- We evaluate them together as an evidence-acquisition tradeoff.

Problem Formulation

A grounded VideoQA example is:

$$x = (v, q, \mathcal{A}, y, \mathcal{T})$$

- v : video, q : question, $\mathcal{A}$: answer options.
- y : correct answer, $\mathcal{T}$: temporal grounding span.
- Candidate evidence pool: $\mathcal{E}(x) = \{e_1, \dots, e_N\}$.

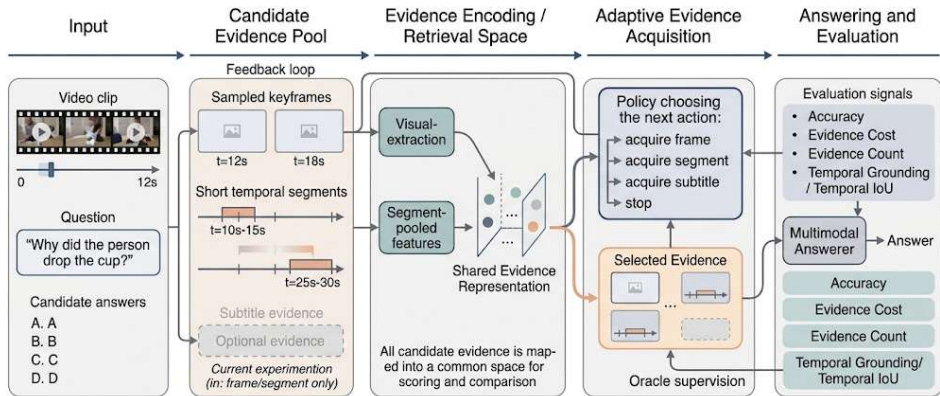
A policy selects evidence $S \subseteq \mathcal{E}(x)$ with cost:

$$C(S) = \sum_{e_i \in S} c_i$$

Goal

Maximize answer accuracy and temporal grounding while minimizing acquired evidence cost.

Method



- Build frame and short segment evidence candidates from NExT-GQA videos.
- Compare one-segment, learned two-item, and fixed-budget evidence policies.
- Re-answer the same evidence using a frozen CLIP-style answerer and Qwen2.5-VL.

Dataset and controls

- NExT-GQA validation: 3,358 examples.
- NExT-GQA test: 5,553 examples.
- Evidence policies:
 - ▶ one segment;
 - ▶ learned two-item policy;
 - ▶ fixed 3 frames + 3 segments.

Metrics

- Acc@QA: answer accuracy.
- Cost and evidence count.
- mIoP, IoP@0.5, mIoU, IoU@0.5.
- Acc@GQA: correct answer and IoP ≥ 0.5 .

Acc@GQA is computed over the selected evidence set.

Result 1: Frozen Answerer Is the Bottleneck

Method	Acc@QA	Cost	IoP@0.5	Acc@GQA
One segment	0.378	1.500	0.311	0.124
Learned two-item policy	0.381	2.795	0.410	0.164
Fixed 3 frames + 3 segments	0.388	7.500	0.615	0.245
Fixed 6 frames + 6 segments	0.394	14.908	0.787	0.316

Takeaway

The learned policy improves grounding, but answer accuracy remains near 0.38 because the frozen answerer is too weak.

Result 2: Stronger Answerer Reveals the Tradeoff

Method	Acc@QA	Cost	IoP@0.5	Acc@GQA
Qwen one segment	0.674	1.500	0.315	0.225
Qwen learned two-item policy	0.695	2.785	0.415	0.302
Qwen fixed 3 frames + 3 segments	0.724	7.500	0.615	0.457

- Learned two-item evidence keeps **96%** of fixed 3+3 answer accuracy.
- It uses only **37%** of the evidence cost.
- It keeps only about **66%** of fixed 3+3 grounded answer accuracy.

Full Test-Set Summary

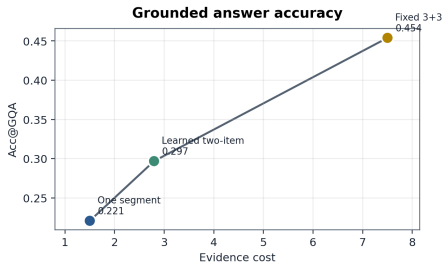
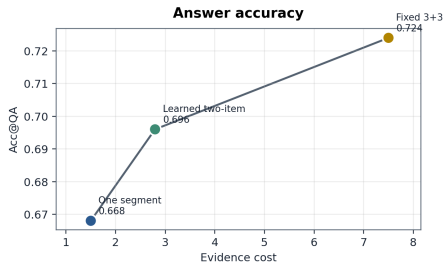
Method	Acc@QA	Cost	IoP@0.5	Acc@GQA
Qwen one segment	0.668	1.500	0.311	0.221
Qwen learned two-item policy	0.696	2.795	0.410	0.297
Qwen fixed 3 frames + 3 segments reference	0.724	7.500	0.615	0.454

Takeaway

The test-set summary preserves the central tradeoff: learned evidence is the better budgeted method, while fixed 3+3 remains the high-coverage grounding reference.

Accuracy-Cost-Grounding Tradeoff

Current Qwen Test Summary: Accuracy-Cost-Grounding Tradeoff



Interpretation

Compact evidence is efficient for answering, but faithful grounding still benefits from more temporal coverage.

Qualitative Lessons

Case	Question	Outcome
Correct and grounded	Why is the boy in yellow reaching out to things on the green mat?	Predicted C, gold C; temporal IoU 0.675
Correct, weakly grounded	Why did the boy pick up one present and move to the sofa?	Predicted C, gold C; temporal IoU 0.047
Grounded but wrong	Why did the man in white hold tightly to the boy in white?	Predicted E, gold D; temporal IoU 0.541

Lesson

Answer correctness and temporal grounding are related but not interchangeable.

Main conclusion

Grounded VideoQA should be evaluated as evidence selection plus answering, not only answer classification.

- A weak answerer hides meaningful evidence-policy differences.
- Qwen2.5-VL makes compact evidence useful for answering.
- Fixed larger evidence still wins when grounding is the priority.
- The final result is an accuracy-cost-grounding tradeoff, not a single winner.

The full test-set summary states the final accuracy-cost-grounding tradeoff used for today's presentation.

Backup: Official Metric Caveat

- NExT-GQA official grounding code expects one predicted temporal span per question.
- Our method selects a set of evidence items.
- The report uses a transparent evidence-set metric: best selected evidence item with $\text{IoP} \geq 0.5$.
- A single-span version requires an aggregation rule such as first selected span, highest-score span, or union span.

How to discuss this in Q&A

We do not claim official single-span leaderboard comparability; we use the metric to study evidence-set support under controlled policies.

Backup: Result Provenance

- Test split size: 5,553 examples.
- The one-segment row comes from the generated Qwen test predictions.
- The learned row is a completed full-test result.
- The fixed 3+3 row is the high-cost coverage reference.

How to discuss this in Q&A

The research conclusion depends on the ordering across operating points and the large cost gap, not on small changes in the final decimal places.